

MATH 213 – Homework 1

In class today, we discussed a graph from the New York Times on COVID vaccination usage in various countries of the world. You can find the document in blackboard. I have removed editing access for the moment, but everyone should be able to view the document.

As a reminder, here are the basics on how the graph was constructed:

- Each bubble represents a country. The size of the bubble corresponds to the country's population.
- Graph is split in two: one piece for below-average vaccination rates and another for above-average vaccination rates.
- Colors correspond to the income-level for each country, grouped into one of four categories – low, lower-middle, upper-middle, and high. The two lower categories are orange and the two upper categories are blue.
- A country appearing to the left of either graph has used a lower percentage of delivered doses. A country appearing the right of either graph has used a higher percentage of delivered doses.

Re-read your group's work on the Google Slides document. Also read the original article in the NY Times, [linked here](#). You will need to create an account to read the article, but you do not need a paid subscription.

1. Quote a sentence or two from the article that describes an aspect of the graph that your group did not mention in the Google Slides document. OR, if you feel that your group covered everything the article does, please choose a sentence from the article and a sentence from your group's work that capture the same idea and paste them here.

"If a country with a low vaccination rate is using a smaller share of the doses it has on hand, it suggests that demand in the country is weak, experts say, or that it lacks the infrastructure to distribute vaccines."

This quote from the article describes an aspect of the graph that our group did not mention in the google slides. While we noticed that some countries in the below average vaccination group are more spread out, we did not explain that this spread could be due to weak demand or infrastructure challenges, as highlighted in the article.

2. Hover over the bubbles in the graph in the online version of the article to see the names of the countries they represent. Keep searching until you find the bubble that represents Ireland. To help you, Ireland is in the bottom right corner of the graph. Write a bulleted list of all of the information you can glean about Ireland and its vaccination program as of December 2021, just by reading this graph. Do not include any information that requires inference or educated guesses, just the facts from the graph.

- Ireland is in the "above-average vaccination rates" section of the graph.
- Ireland has used 87% of the doses delivered to it.
- Ireland is represented as a high-income country (blue bubble)
- Irelands vaccination rate is above the global average of 45% as of December 2021

3. Find the graph in the [article](#) titled "Wealthier countries' head start on vaccination has kept them in the lead." (The one with the floating bubbles.) Notice that there is a play button on the top right corner of the graph that you can use to replay the animation.

Describe what is going on in this graph:

- a. Describe what is going on in the graph at the beginning before the bubbles start to move.

At the beginning of the timeline (January 1, 2021 – January 10th, 2021), most countries have very low vaccination rates, as the global rollout of vaccines had just started. Wealthier countries (blue bubbles) are clustered towards the right side of the graph, indicating higher GDP per capita, but their vaccination rates are still low. On the other hand, lower income countries (orange bubbles) are clustered towards the left with even lower vaccination rates.

- b. Describe what starts to happen in the graph when you press the “play” button. In other words, what is the significance of the rising bubbles? (Make sure to write in full sentences and to speak in context – talk about vaccination rates, countries, income levels.)

After clicking the play button, I noticed that the bubbles (both blue and orange) begin to move higher, indicating an increase in vaccinations for their populations. Additionally, I observed that wealthier countries (represented by blue bubbles) are rising noticeably faster than lower-income countries (represented by orange bubbles). This shows that high-income (upper-middle / high) countries are vaccinating their populations at a much quicker pace compared to lower-income countries.

- c. Stop the graph in around July 2021. What story does the graph tell you at this point? (2-3 sentences is fine. Write in full sentences. Again, speak in context.)

By July 2021, the graph shows a clear separation between the high-income and low-income countries in terms of vaccination rates. Wealthier countries (blue bubbles) such as the United States, Chile and Hungary have significantly higher vaccination rates, with many administering over 100 doses per 100 people. In contrast lower-income countries (orange bubbles) such as Mozambique and Mali have much lower vaccination rates often below 50 doses per 100 people.

- d. Let the animation run until it ends. What story does the graph tell you now? (2-3 sentences is fine. Write in full sentences. Again, speak in context.)

After running animation all the way through (December 8, 2021) the graph shows that the gap between high income and low-income countries have increased even further. Wealthier countries (blue bubbles) have achieved extremely high vaccination rates, with many administering over 200 doses per 100 people. In contrast the lower income countries (orange bubbles) continue to lag with vaccination rates below 50 doses per 100 people. This shows the global inequality in vaccine distribution, wealthier countries maintaining a significant advantage over lower-income countries

4. Lastly, please watch the following four short videos on effective group work and write a couple of sentences answering the three questions that follow. We'll discuss your answers as a group and as a class on Friday.

- [Video 1](#)
- [Video 2](#)
- [Video 3](#)
- [Video 4](#)

- a. **In both Videos 1 and 2, one of the partners was making group work difficult. Describe in your own words the behavior of the partner who was creating challenges.**

In both videos 1 and 2, one of the partners was making group work difficult. In video 1, I noticed that the male in the blue shirt was causing issues because he was not collaborating effectively with his partner or offering any helpful insights into the problem at hand. In video 2, the male wearing the black shirt was the one creating challenges within the group. He was rude and disrespectful in response to a very normal request in the group.

- b. **Videos 3 and 4 show examples of partners working well together on a task. Describe in your own words the strategies that the team members are using that create a positive atmosphere for effective learning.**

Videos 3 and 4 show examples of partners working well together on a task. In both videos, the partners demonstrate effective strategies such as open communication, asking relevant questions, and supporting each other. For example, in video 3 when the person on the right did not understand something, their partner was happy to explain it to them. This demonstrates the strategies mentioned above and creates a positive atmosphere for effective learning.

- c. **As we have discussed in class already, we will be striving to work effectively in groups this semester. Describe in your own words your experiences with group work in your past academic life. Make sure to note any negative experiences (explain why those experiences were negative to the extent to which you are comfortable sharing with the class) and also include any positive experiences, if you have had any.**

On the positive side, I've worked in groups where everyone communicated openly, shared ideas and divided tasks fairly. For example, in one project my group members and I assigned roles based on each person's strengths which made the work a lot more enjoyable. We also had a group chat so when someone was struggling, they could reach out for help at any time without waiting for the next in-person meeting.

On the negative side, I have also been in groups where members did not contribute equally. In one instance, a group member only took notes on what I and the other person did and did not actually complete any work for the project. As a result, most of the workload fell on me and my other teammates, which put extra pressure on us to complete more than we originally planned for.